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(57) Abstract:

Abstract: FIELD-EYE operates as an artificial intelligence agricultural robot which performs real-time plant health assessments and precise pest management operations throughout small and medium-sized agricultural fields. The system combines an ESP32-CAM module with motor control components and ultrasonic sensors and soil sensors and GPS tracking systems and a pesticide spraying device to create a cost-effective power-efficient robotic system. The rover uses autonomous operation tomove through agricultural fields which researchers divided into grid sections while it takes high- resolution crop pictures and runs machine learning models that detect pests and diseases andnutrient deficiencies through edge computation. The system uses on-device intelligence and cloud- based dashboards to provide early detection and accurate location mapping and reduced dependence on human labor and data-based agricultural decision-making. The invention is capable of functioning in both autonomous and semi-autonomous modes, enabling complete automated navigation and spraying or manual control with AI suggestions. The multilingual advisory module also improves the usability of the invention by providing real-time notifications for alerts, crop analysis, and spraying notifications in multiple regional languages through mobile and web interfaces. The invention has dual connectivity options (Bluetooth and Wi- Fi/GSM/LoRa), hardware expansion capabilities, and solar assistance for power, making FIELD- EYE an affordable, scalable, and intelligent smart farming solution for inclusive precision agriculture.

FIELDEYE-AI driven plant health monitoring system

Abstract:

FIELD-EYE operates as an artificial intelligence agricultural robot which performs real-time plant health assessments and precise pest management operations throughout small and medium-sized agricultural fields. The system combines an ESP32-CAM module with motor control components and ultrasonic sensors and soil sensors and GPS tracking systems and a pesticide spraying device to create a cost-effective power-efficient robotic system. The rover uses autonomous operation tomove through agricultural fields which researchers divided into grid sections while it takes high-resolution crop pictures and runs machine learning models that detect pests and diseases andnutrient deficiencies through edge computation. The system uses on-device intelligence and cloud- based dashboards to provide early detection and accurate location mapping and reduced dependence on human labor and data-based agricultural decision-making.

The invention is capable of functioning in both autonomous and semi-autonomous modes, enabling complete automated navigation and spraying or manual control with AI suggestions. The multilingual advisory module also improves the usability of the invention by providing real-time notifications for alerts, crop analysis, and spraying notifications in multiple regional languages through mobile and web interfaces. The invention has dual connectivity options (Bluetooth and Wi-Fi/GSM/LoRa), hardware expansion capabilities, and solar assistance for power, making FIELD-EYE an affordable, scalable, and intelligent smart farming solution for inclusive precision agriculture.

Description:

Background of the Invention:

The traditional method of monitoring crops in agriculture relies on manual inspections, which take a lot of time and aren't very efficient, especially in large fields. This makes it hard to spot early infestations of pests, plant diseases, and nutrient deficiencies. Current automated solutions usually require expensive processing hardware and constant cloud connectivity, which makes them too costly and unreliable for small and medium farmers, particularly in rural areas with poor network access. Additionally, these solutions often do not provide real-time decision support, flexible operation control, or multilingual advice for farmers with limited technical skills. Therefore, there is a need for a low-cost, AI-based agricultural rover that offers autonomous and semi-autonomous operation, real-time crop health monitoring, precise interventions, edge intelligence, and effective communication with farmers.

Summary of the Invention:

The current invention offers FIELD-EYE, an agricultural rover that uses AI technology for real-time crop health observation and precise pest control. The invention comprises an ESP32-CAM microcontroller, motor control system, ultrasonic and soil sensors, GPS location tracking, pesticide spraying system, and communication systems that use both Bluetooth and Wi-Fi technology. The agricultural rover is capable of moving through grid-divided cropfields, taking high-resolution images of the crops, and running machine learning algorithms on the rover to observe pests, diseases, and nutrient deficiencies in the crops with minimal latency and reliance on cloud computing. The invention operates in two modes which include complete autonomy and partial autonomy. The autonomous mode of the rover enables automatic grid movement while it uses AI technology to execute crop observation and analysis and precise pesticide spraying. The semi-autonomous mode allows users to control the rover manually while they use AI crop analysis recommendations. The system includes an advisory system which operates in multiple languages to deliver real-time crop health assessments and spraying alerts and diagnostic information through both mobile and web platforms.

Claims An AI-based agricultural rover for plant health monitoring and pest control, the system including a microcontroller unit with an integrated camera module, motor control system, navigation system, communication interface, and an artificial intelligence model integrated onboard and trained for crop analysis. The agricultural rover as claimed in claim1,where in the microcontroller is an ESP32-CAM designed to capture images of plants and run machine learning models on the device for real-time detection of pests, plant diseases, and nutrient deficiencies. The agricultural rover as claimed in claim 1 operates through its complete autonomous system which enables grid-based field navigation and automatic image acquisition and artificial intelligence decision-making and precise pesticide spraying operations. The invention provides an affordable solution which can be expanded through its data-driven smart farming system that uses edge intelligence and modular hardware design and farmer advisory support.

The agricultural rover as claimed in claim 1, which includes a system that enables two operational modes, which allow users to control navigation through manual methods while the system provides guidance through artificial intelligence and users can take control whenever they want. The agricultural rover as claimed in claim 1, which has a multilingual advisory module that creates real-time alerts and crop health diagnostics and spraying notifications in multiple regional languages through a mobile or web-based dashboard. The agricultural rover as claimed in claim 1, which includes soil sensors and ultrasonic sensors and GPS tracking modules, enables users to detect obstacles and monitor soil conditions and create pest maps based on their current location. The agricultural rover as claimed in claim 1 contains a communication interface which uses Bluetooth technology together with at least one of Wi-Fi or GSM or LoRa modules that enable users to monitor activities and transmit data over distances.

Brief Description of the Drawings

FIG. 1 is a detailed schematic circuit diagram showing the integrated electronic control system, power distribution network, and dual microcontroller design of the autonomous agricultural rover.
FIG. 2 is an operational flowchart showing the decision-making logic of the AI system, illustrating the interaction between autonomous scanning, pest detection, and the farmer approval feedback loop.

Description of Preferred Embodiment

In FIG. 1, the 12V Power Source is the central energy source for the entire system. The high-voltage bus is directly connected to the L298N Motor Driver for high-current actuation of mechanical motion. At the same time, voltage regulation is achieved through a 5V BuckConverter to supply a stable low-voltage logic signal to the microcontrollers, safeguarding the digital circuitry from electrical spikes.
The system design employs a split processing framework to distinguish between extensive image processing and real-time navigation:

Vision and AI Module (ESP32-CAM): The ESP32-CAM is the central image acquisition module. As depicted in the system design, it is carefully connected to acquire real-time data from the field. This module implements an AI model trained on a proprietary image dataset of 11,000 images to detect pests, weeds, and plant diseases.
Navigation and Telemetry Module (ESP8266): The ESP8266 Micro-controller is the central logic module for rover motion control and IoT communication. It is connected to the motor driver to control the motion of Motor 1, Motor 2, Motor 3, and Motor 4.
Actuation Interface (L298N): The L298N Motor Driver accepts digital inputs from the ESP8266 to control the four-motor motion system, facilitating high-torque motion control and positioning in the field.

The operational logic of the system is further explained in FIG.2. The process begins with the rover scanning the agricultural field in a structured grid pattern. Once an image is taken at a particular cell grid, the ESP32-CAM sends the image and location information to the user's mobile device. The AI module determines the classification of the presence of pests and diseases.

After the classification, the system proceeds with either of the two operational modes:
Semi-Autonomous Mode: The system proceeds with a query to request Farmer’s approval to spray pesticides. If "Yes" is received, the rover proceeds with a targeted spray; otherwise, it moves on to the next grid.
Autonomous Mode: After positive classification, the system skips the approval step and proceeds with an immediate targeted spray or weeding operation.

Claims

1. An AI-based agricultural rover for plant health monitoring and pest control, comprising a microcontroller unit with an integrated camera module, motor control system, navigation system, communication interface, and an onboard artificial intelligence model designed for crop analysis.

2. The agricultural rover according to claim 1, wherein a ESP32-CAM micro-controller has been provided to allow images of crops to be taken, from which machine learning algorithms can be executed to identify pests, plant diseases, and nutrient deficiencies, all while in real-time.

3. The agribot as recited in claim 1 has been built so that it operates automatically, with grid navigation across the field, automatically capturing images (with no human intervention), making AI based decisions and precisely spraying pesticides.

4. The agricultural rover as claimed in claim 1, wherein the system is intended to operate in a semi-autonomous manner that allows for manual control of the navigation system with AI suggestions.

5. The agricultural rover as claimed in claim 1, wherein the system further includes a multilingual advisory system that is capable of providing real-time notifications, crop health analysis, and spraying alerts in various regional languages of choice via a mobile or web-based interface.

6. The agricultural rover as claimed in claim 1, wherein the system further includes soil sensors, ultrasonic sensors, and GPS tracking systems that are capable of facilitating obstacle detection, soil analysis, and location-specific pest mapping.

7. In the agricultural rover described in claim 1, there is a communication interface containing Bluetooth and/or one of Wi-Fi, GSM and/or LoRa modules to allow remote monitoring and data exchange.

8. The agricultural rover as detailed in claim 1 comprises a modular design that can accommodate additional hardware components for purposes such as spraying, soil sensing, weed elimination, etc

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